



Global Potato Production Trends: Strategic Insights for Germany's Agricultural Cooperation

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Executive Summary

This analysis of global potato production trends, based on FAO data, identifies the key players shaping agricultural supply chains and can give insights that can support decision-making of stakeholders.

Key Findings:

- ☐ China, India, and Russian Federation dominate overall output, while New Zealand and USA stand out as an efficiency leader.
- ☐ Global production has grown steadily since the 1990s, though risks remain in volatile regions such as Eritrea and Botswana.

Recommendations:

- ☐ Looking forward, Germany should maintain strong ties with established producers.
- ☐ Besides, Germany should actively engage with emerging growth countries in Africa and the Middle East, including Guinea, Senegal, Kenya, and Iraq. These partnerships would not only diversify supply sources but also align with Germany's strategic goal of promoting sustainable global food security.



Introduction

In this report, I present the findings of data analysis on global potato production trends and is to be issued to the stakeholders. The objective was to transform raw agricultural production data from UN's Food and Agriculture Organization (FAO) into meaningful insights that can support decision-making in the context of food security, agricultural trade and international cooperation.

This report addresses several issues about potato production, including identification of the world's leading producers, yield efficiency among the countries, volatility risks, long-term growth patterns and distribution inequality. We applied SQL queries, statistical calculations, and data visualization techniques to explore both global trends and to compare the trends in Germany with the potential cooperation partner United States of America.

Methodology

Data

The dataset in this report was provided by FactSet, that was collected from UN's Food and Agriculture Organization (FAO). This dataset contains production data for 205 food items across 211 countries between 1961 and 2023. This report indeed focusses on the food item *Potatoes* and covers the years between 2000 and 2023 unless indicated. The measures that are considered in the analysis for the food item are summarized in the Table 1.

Element	Unit
Area harvested	ha
Production	t
Yield	kg/ha

Table 1: FAO measures available for Potatoes with units.

Methods

To conduct the analysis, a combination of data and visualization tools were used:

- ☐ SQL - Used to query the FAO database, clean the duplicated data (such as China and territories) to ensure consistency and accuracy, and perform aggregations and filters for each analytical question.
- ☐ Power BI – Used to transform query results from SQL into clear and interactive visualizations.
- ☐ Microsoft Excel - Used to transform query results from SQL into basic visualizations.

Analysis & Findings

Global Leaders & Champions

Leading Potatoes Producers

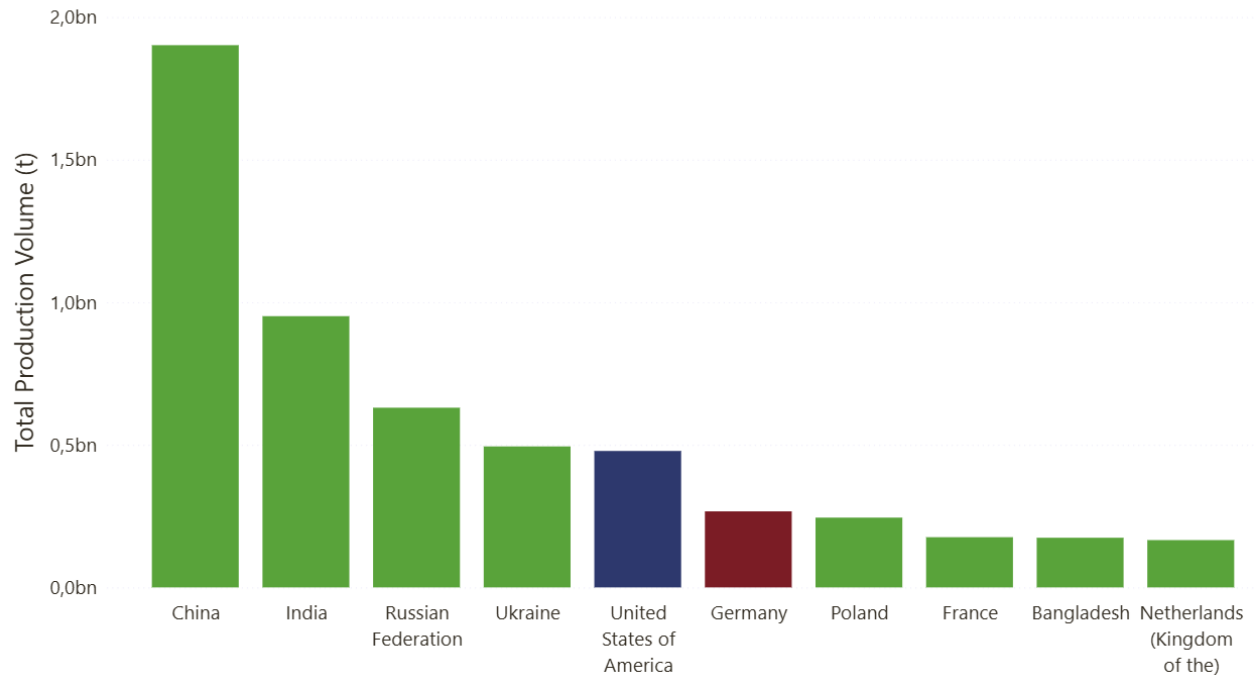


Figure 1. Top 10 global producers of Potatoes between 2000 and 2023. The chart shows production volumes in metric tons, highlighting the production of United States of America (blue) and Germany (red).

Yield Champions



Figure 2. Yield vs. harvested area for Potatoes, 2000-2023. Each point represents a country, with bubble size indicating production volume. High-yield countries can be identified regardless of land area cultivated. United States of America (blue) and Germany (red) are highlighted.

Yearly Champions

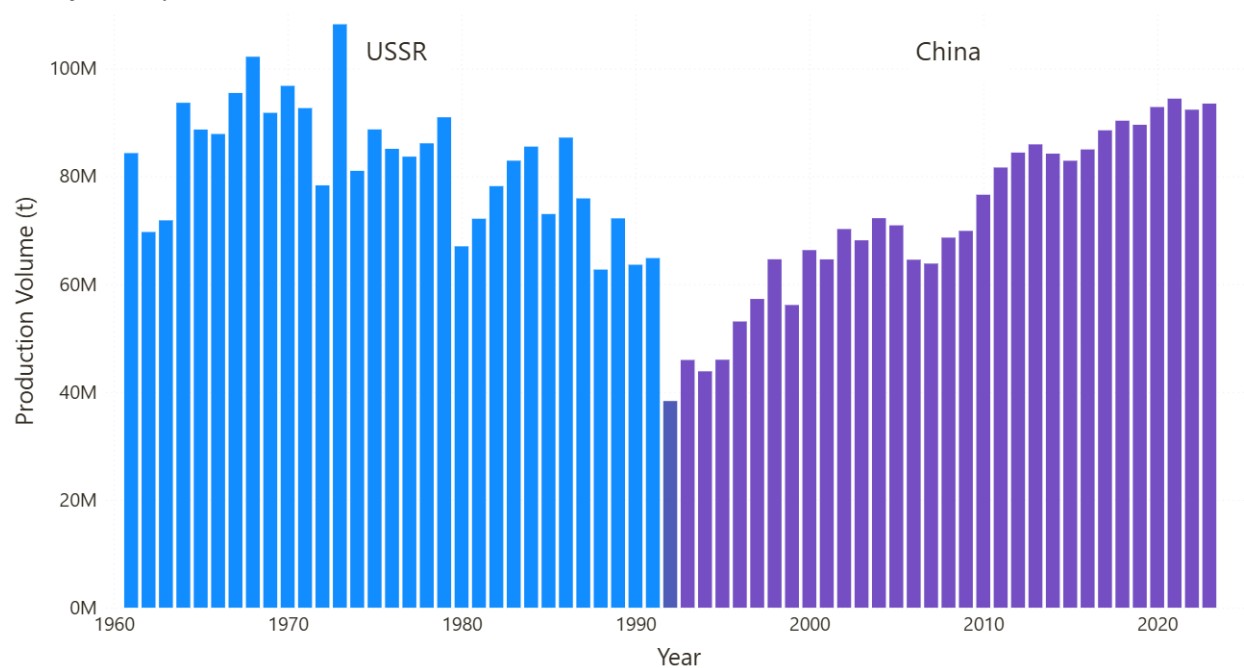


Figure 3. Annual leaders in Potato production, 1961-2023. The bars show which countries held the top producer position in each year. USSR, Russian Federation and China are shown in blue, navy and purple, respectively.

Trends & Changes over Time

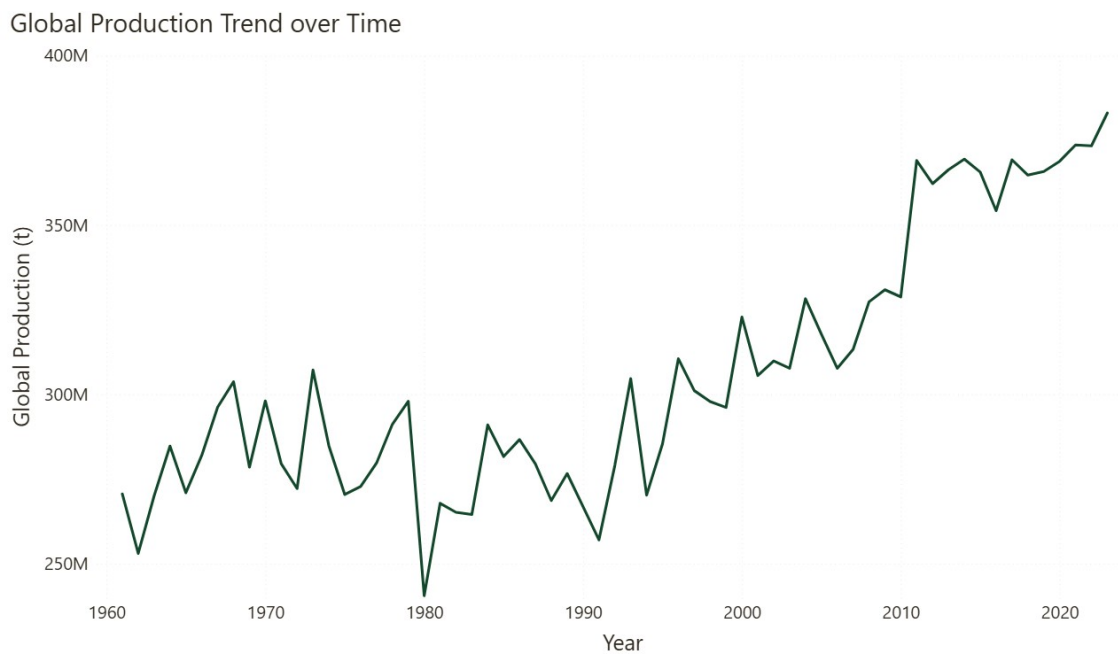


Figure 4. Global Potato production trends, 1961–2023. Annual production volumes illustrate the constant trend until 1991 with notable dips linked to external shocks and long-term growth trajectory after 1991.

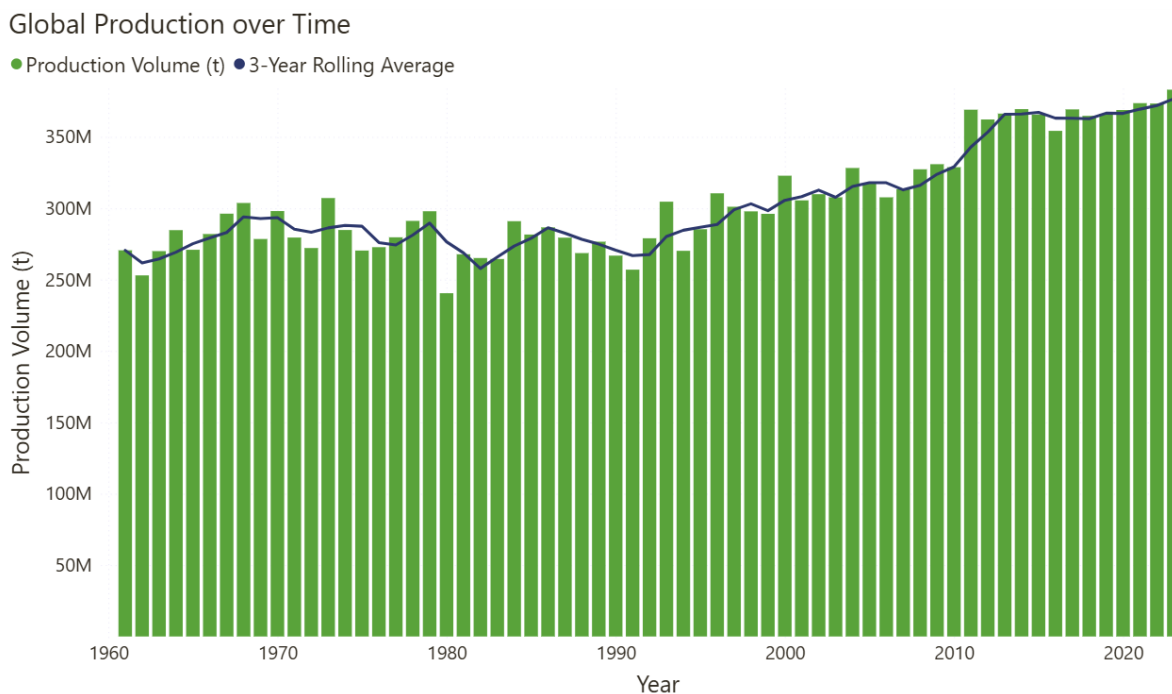


Figure 5. Smoothed production trends using three-year rolling averages globally, 1961–2023. This highlights underlying patterns beyond short-term fluctuations.

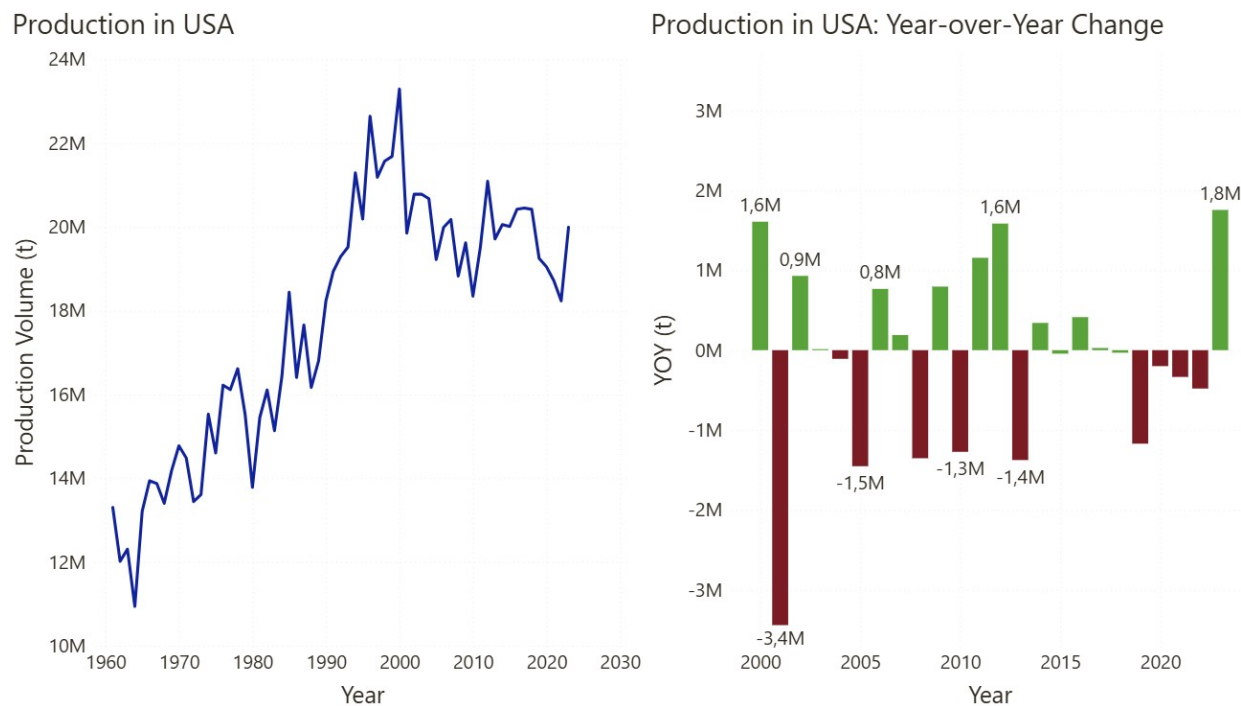


Figure 6. USA's production journey, 1961–2023 (left). The bar chart tracks year-over-year change (YOY) between 2000 and 2023 (right).

Most Improved Countries over Last 5 Years

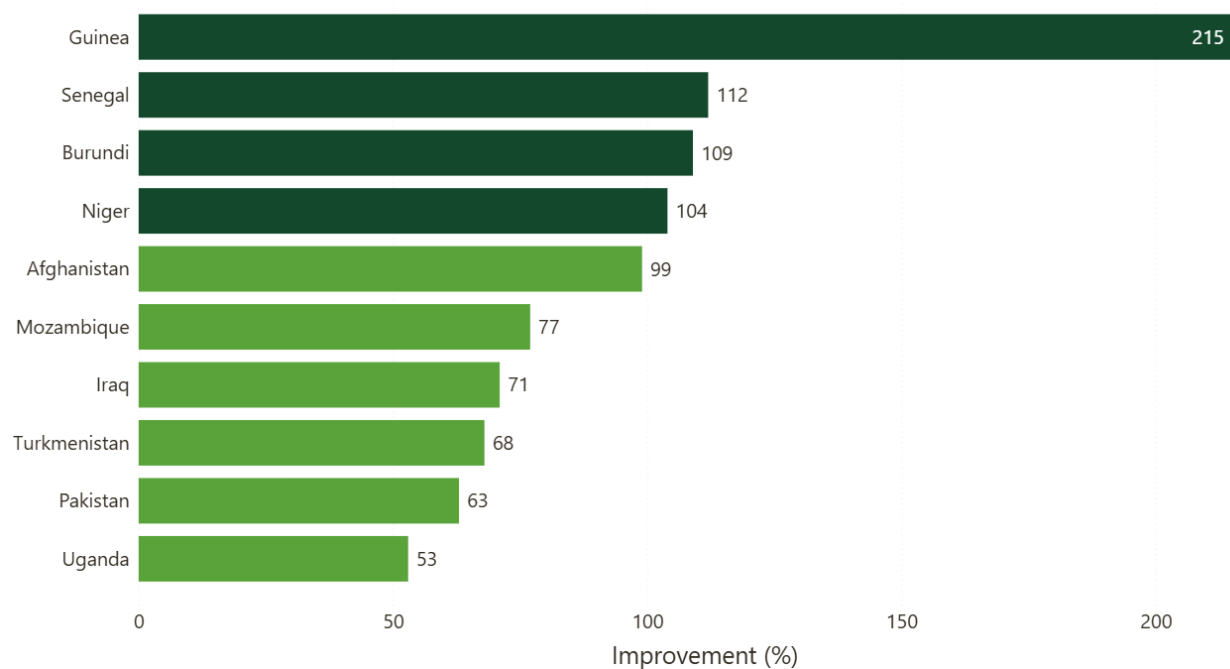


Figure 7. Five-year improvements in Potato production, from 2014–2018 to 2017–2023. Countries with the growth greater than 100% in production volumes are highlighted.

Risk & Volatility

Yield Volatility

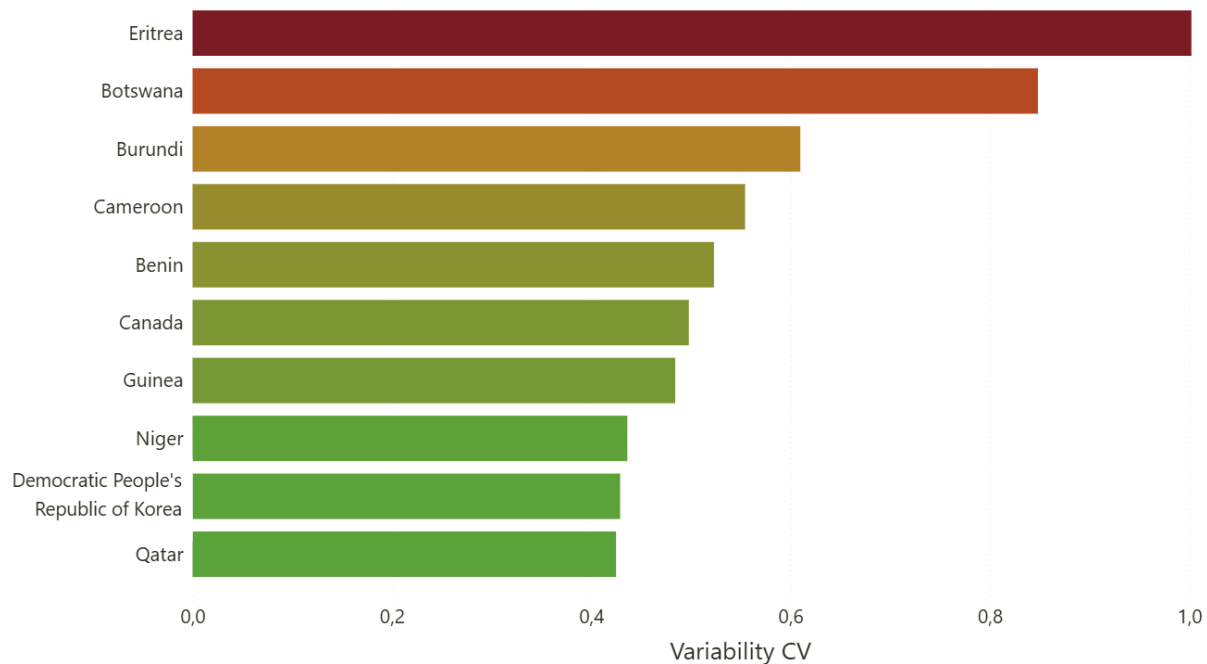


Figure 8. Top 10 countries with the highest yield variability coefficient value (CV), 2000–2023.

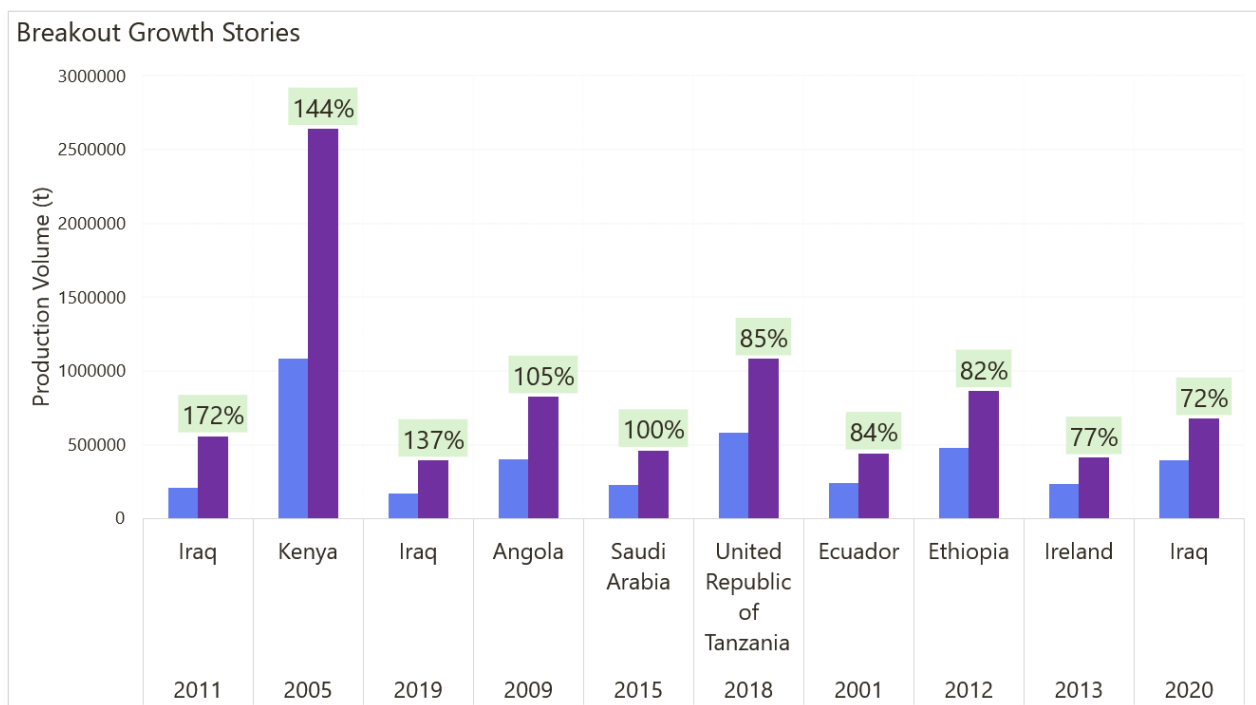


Figure 9. Highest year-on-year growth rates in Potato production by country, 2000–2023. Spikes highlight sudden increases in output, signaling potential breakout growth stories.

Global Distribution & Inequality

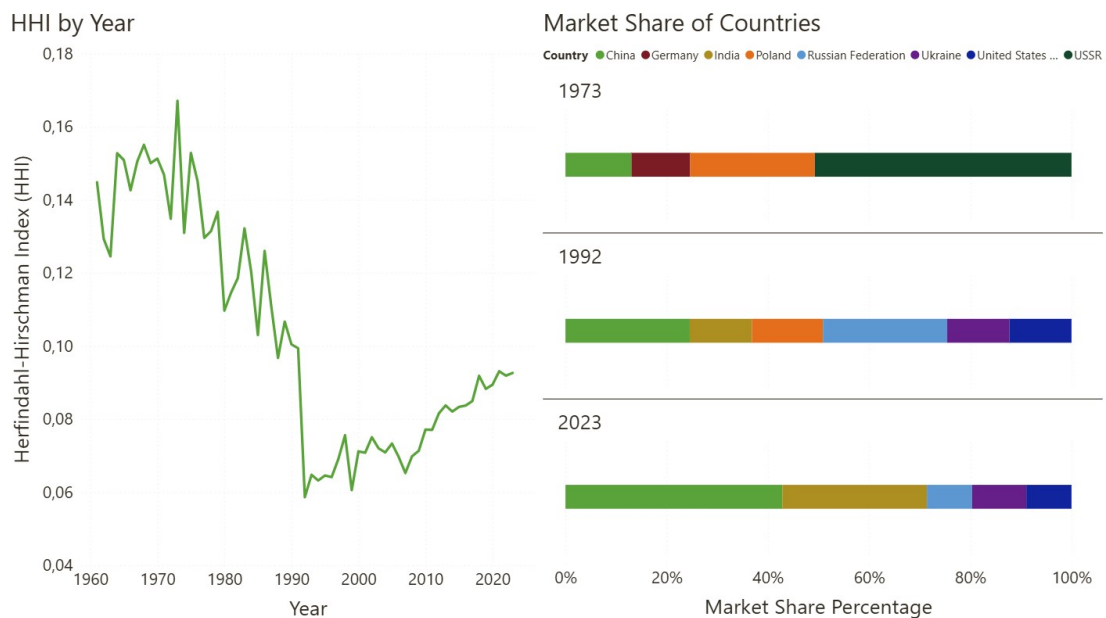


Figure 10. Inequality in global Potato production shown by Herfindahl-Hirschman Index (HHI) (left). The bar chart shows market share of leading countries (>5%) in the years with maximum and minimum HHI (1973, 1991), as well as the latest year (2023).

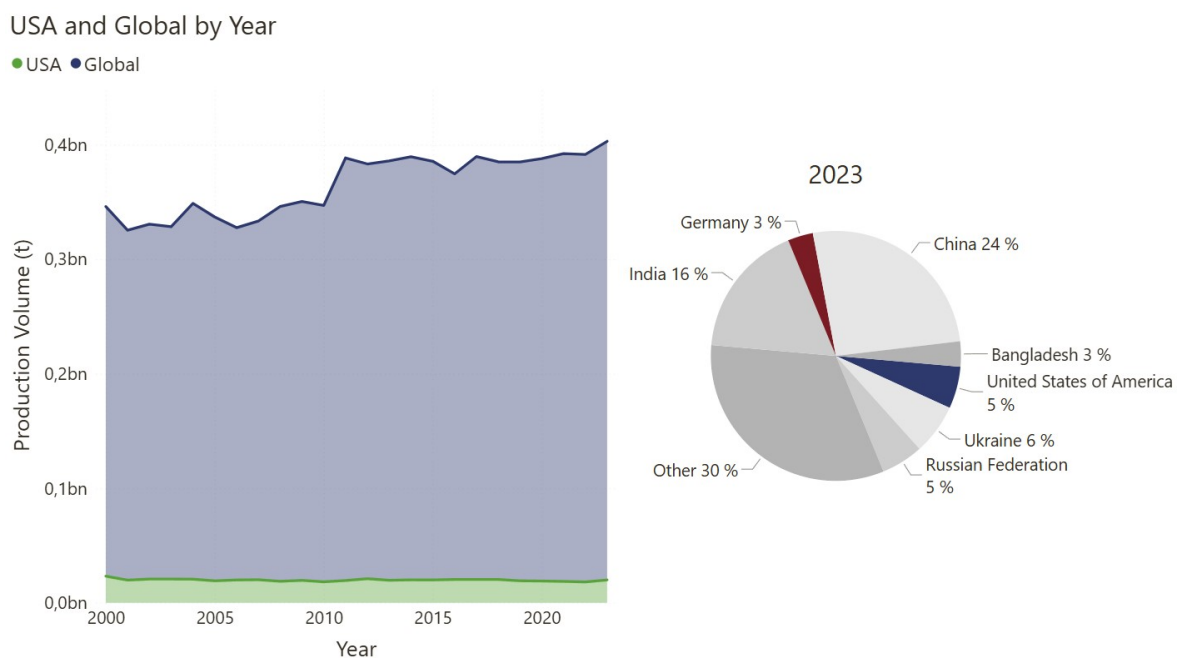


Figure 11. Share of global Potato production contributed by USA and Germany. The stacked area chart compares USA's share with global volume, 2000–2023 (left). The pie chart highlights the market share of Germany (red) and USA (blue) compared to rest of the world in 2023 (right).



Results

The analysis revealed that China, India, and the Russian Federation dominate global production of potatoes, together supplying the largest share of total output (Figure 1). In terms of efficiency, New Zealand achieves the highest yield per hectare, although the United States produces 40 times more in volume with only a modestly lower yield (Figure 2).

Global production remained relatively constant between 1961 and 1991, before entering a sustained growth phase after 1992 (Figure 4). Notable downward spikes occurred in 1962, 1980, and 1991, while an upward surge was observed in 2011. The downward spikes can be explained by the adverse weather events (droughts or frost) combined with economic or political instability—especially around the collapse of the Soviet Union in 1991, which disrupted farming and distribution in large producing regions. The upward spike around 2011 likely results from favorable growing seasons in key countries, improved agronomic practices, and renewed investment in agriculture following earlier downturns.

Yield volatility was most pronounced in Eritrea and Botswana, indicating unstable production trends (Figure 8). These two countries with frequent or extreme swings illustrate higher risk in supply stability.

Several countries experienced breakout growth, most notably Iraq (+172% in 2011; +137% in 2019), Kenya (+144% in 2005), Angola (+105% in 2009), and Saudi Arabia (+100% in 2015), this could be due to the technology adoption, infrastructure improvements, and expanded cultivation, often following periods of relative political stability (Figure 9).

Production varies from moderately concentrated to unconcentrated among years, as confirmed by the Herfindahl–Hirschman Index (HHI), which ranged from 0.17 in 1973 to 0.06 in 1992 (Figure 10). Before 1992, the USSR was the leading producer, but China has since assumed market leadership (Figure 3).

USA showed steady growth from 1961 to 1992, followed by a stable plateau with periodic spikes, on the contrary to global production trend (Figure 6). In 2023, USA contributed 5% to the global output, while Germany accounted for 3% (Figure 11). Recent five-year growth leaders included Guinea (+200%), Senegal, Burundi, and Niger (greater than 100%), highlighting the rapid expansion of smaller producers in Africa (Figure 7).



Conclusion

This analysis highlights the shifting dynamics of global potato production over the past six decades. While China India and Russian Federation dominate in terms of volume, efficiency leaders such as New Zealand and USA demonstrate the importance of sustainable yield management. Long-term growth since the 1990s contrasts with episodes of volatility driven by climate shocks, political upheavals, and economic change.

Emerging producers in Africa and the Middle East—including Guinea, Senegal, Kenya, and Iraq—signal new opportunities for agricultural cooperation, though their growth is often accompanied by instability. For Germany, the strategic path forward lies in balancing partnerships with established producers while fostering collaboration with high-potential newcomers, both to diversify supply chains and to advance global food security goals.

The results point to four main directions for German agricultural cooperation strategy:

- ❑ **Secure Core Partnerships:** Continue close engagement with China, India, and USA, which collectively account for the bulk of global output.
- ❑ **Leverage Knowledge Transfer:** Collaborate with efficiency leaders like New Zealand to gain insights into sustainable yield practices that can be adapted to German contexts.
- ❑ **Diversify with Emerging Producers:** Explore partnerships with high-growth countries in Africa and the Middle East (e.g., Guinea, Senegal, Iraq, Kenya). While these markets are less stable, they represent opportunities to strengthen global food security and reduce dependence on a few giants.
- ❑ **Risk Monitoring:** Closely watch volatile producers (e.g., Eritrea, Botswana) and track shifts in global leadership to anticipate future disruptions.